

How far is a given function of two real variables from factorized form?

Consider a given $f(x, y)$. How could one quantify how far it is from factorized form $g(x)h(y)$? For this question to make sense, one first has to find an optimal (nearest) pair of g and h . That is a variational problem. A standard way of defining distance between functions is the root mean square difference (L_2 norm). So, one has to find an extremum of the mean square difference

$$J = \int dx dy (f(x, y) - g(x)h(y))^2 .$$

Variation with respect to $g(x)$ and $h(y)$ yields

$$\begin{aligned} 0 &= \frac{\delta J}{\delta g(x)} = \int dy f(x, y)h(y) - g(x) \int dy h^2(y) \\ 0 &= \frac{\delta J}{\delta h(y)} = \int dx f(x, y)g(x) - h(y) \int dx g^2(x) \end{aligned}$$

which can be solved for

$$h(y) = \frac{\int dx f(x, y)g(x)}{\int dx g^2(x)} ,$$

which can then in turn be reinserted to eliminate $h(y)$ and yield

$$g(x) = \left[\int dy \left[\frac{\int dx f(x, y)g(x)}{\int dx g^2(x)} \right]^2 \right]^{-1} \int dy f(x, y) \frac{\int dx f(x, y)g(x)}{\int dx g^2(x)}$$

This nonlinear equation is amenable to iterative numerical solution. Inserting an initial guess for g on the right hand side ($g = 1$ appears to work in practice), the left hand side constitutes a refined form for g , which can again be reinserted on the right hand side, etc., until convergence. Convergence appears to be quite rapid in practice (5-10 iterations).

Having found an optimal pair of g and h , it makes sense to compare the L_2 norm separating $f(x, y)$ from $g(x)h(y)$, i.e., $\sqrt{J}$, with the L_2 norm of f itself. I.e., denoting $N = \int dx dy f^2(x, y)$,

$$\text{Relative distance of } f \text{ from factorized form} = \sqrt{\frac{J}{N}}$$